Divergent Tobacco Tax Policies and Their Impact on State Tax Revenues and Small Retail Businesses: A Comparative Analysis of Virginia and Maryland, 2021–2025

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ABSTRACT

This study examines the fiscal and economic consequences of divergent tobacco tax and regulatory policies in Virginia and Maryland between fiscal years 2021 and 2025. Using annual revenue data drawn directly from the Virginia Department of Taxation Annual Reports and the Maryland Alcohol, Tobacco, and Cannabis Commission (ATCC) Annual Reports, this paper analyzes how contrasting policy regimes — Virginia maintaining a stable $0.60 per pack cigarette tax with no statewide flavor ban, versus Maryland raising its cigarette tax from $2.00 to $5.00 per pack and imposing aggressive product regulations — have produced measurably different fiscal outcomes for each state. Key findings include: (1) Virginia's cigarette tax revenue declined by 33.5 percent over the study period ($234.2M in FY2021 to $155.8M in FY2025), driven primarily by long-run consumption decline rather than tax policy; (2) Maryland's cigarette stamp revenue followed a more volatile trajectory, surging 29.7 percent in FY2022 following its tax increase before declining sharply by 12.0 percent in FY2024 and partially recovering in FY2025; (3) Other Tobacco Products (OTP) revenue grew consistently in both states, confirming a product substitution trend rather than overall tobacco abandonment; and (4) the persistent $4.40 per pack tax differential between the two states creates a structural economic incentive for cross-border purchasing that may be suppressing Maryland's revenue potential. An interactive visualization dashboard: → View Dashboard

Keywords: *tobacco tax policy, cigarette excise tax, state tax revenue, cross-border purchasing, OTP substitution, Virginia, Maryland, fiscal policy, retail business impact, data visualization*

INTRODUCTION

Tobacco taxation sits at the intersection of public health policy, fiscal management, and small business economics. In the United States, each state exercises independent authority to set tobacco excise tax rates, define taxable product categories, and impose regulatory restrictions — a patchwork of policies that generates measurable variation in fiscal outcomes, consumer behavior, and business conditions across state lines. Nowhere is this divergence more striking than in the mid-Atlantic region, where Virginia and Maryland have pursued fundamentally different tobacco policy trajectories over the past five years.

Virginia has maintained a relatively stable tobacco tax environment, keeping its cigarette tax at $0.60 per pack — one of the lowest in the nation — while implementing modest structural reforms to its Other Tobacco Products (OTP) and liquid nicotine tax framework. Maryland, by contrast, has pursued an aggressive tax escalation strategy, raising its cigarette tax from $2.00 per pack in 2020 to $3.75 in March 2021, and again to $5.00 in July 2024 — a 150 percent increase over four years. Maryland has also imposed some of the strictest electronic smoking device regulations on the East Coast and has repeatedly debated comprehensive statewide flavor bans.

These divergent policy choices present a natural quasi-experimental setting for studying how tobacco tax policy affects state fiscal revenues and small retail businesses. With a combined cigarette tax gap reaching $4.40 per pack by fiscal year 2024, the two neighboring states offer a rare opportunity to observe the behavioral and fiscal consequences of dramatically different regulatory regimes within a geographically contiguous area where cross-border consumer movement is both economically rational and physically easy.

This paper addresses that gap by constructing a five-year annual dataset of tobacco tax revenue collections for Virginia and Maryland, sourced directly from each state's official

government annual reports, and analyzing the fiscal, behavioral, and economic implications of the policy divergence observed between 2021 and 2025. An interactive data visualization dashboard presenting all findings is publicly available at: → View Tableau Dashboard

LITERATURE REVIEW

Tobacco Tax Policy and State Revenue

The economic literature on tobacco taxation is extensive, spanning public health, public finance, and behavioral economics. The foundational insight — that tobacco taxes reduce consumption through price effects — is well established (Chaloupka & Warner, 2000). Meta-analyses of price elasticity studies consistently find that a 10 percent increase in cigarette prices reduces consumption by approximately 3 to 5 percent among adults and 7 to 11 percent among youth (Gallet & List, 2003; IARC, 2011). These findings have supported tobacco tax increases as a dual-purpose instrument: simultaneously discouraging use and generating revenue.

However, the fiscal arithmetic of tobacco taxation is more complicated than the public health framing suggests. Because tobacco taxes are specific excise taxes applied to a shrinking consumption base, revenue projections must account not only for price elasticity but also for long-run secular trends in smoking prevalence, cross-border purchasing behavior, product substitution, and illicit trade. The Tax Foundation has consistently documented that states with high tobacco taxes tend to generate significant smuggling inflows from lower-tax neighbors, with some states importing most of their consumed cigarettes from outside the tax jurisdiction (LaFaive & Nesbit, 2023).

Cross-Border Purchasing and Tax Arbitrage

When adjacent jurisdictions impose significantly different tobacco tax rates, economically rational consumers have an incentive to purchase in the lower-tax jurisdiction. This cross-border purchasing behavior has been documented extensively in the literature. Hyland et al. (2005) found that smokers in high-tax states were significantly more likely to report purchasing cigarettes in lower-tax jurisdictions. Lovenheim (2008) used individual-level data to demonstrate that cross-border shopping reduces cigarette tax revenues by a measurable amount in high-tax states proximate to low-tax borders.

The Tax Foundation's annual cigarette smuggling analysis consistently ranks mid-Atlantic and northeastern states among the highest net importers of smuggled and cross-border cigarettes (Hoffer & Macumber-Rosin, 2023). Maryland has historically appeared in this analysis as a significant net importer, a finding consistent with its high tax rate relative to neighboring Virginia.

Product Substitution and the OTP Revenue Base

A significant development in tobacco economics over the past decade has been the rapid growth of non-cigarette tobacco and nicotine products. The economic literature has documented a substitution relationship between cigarettes and electronic nicotine delivery systems: as cigarette prices rise, some consumers shift to ENDS rather than quitting (Cotti et al., 2020; Pesko et al., 2020). The Massachusetts flavor ban of 2020 provided a widely-studied natural experiment — post-ban data revealed that menthol cigarette sales in neighboring states increased by approximately 70 percent in the first three months (Orzechowski & Walker, 2020).

METHODOLOGY

Research Design

This study employs a comparative longitudinal design, analyzing annual tobacco tax revenue collections in Virginia and Maryland across five fiscal years (FY2021 through FY2025). Virginia serves as a low-tax, low-regulation comparison state, while Maryland serves as the high-tax, high-regulation treatment state.

Data Sources

All revenue data was obtained directly from official government sources. Virginia data was obtained from the Virginia Department of Taxation Annual Report for Fiscal Year 2025, specifically Table 5.2: Other Taxes Net Revenue Collections — Non-General Fund. Maryland data was obtained from the Maryland Alcohol, Tobacco, and Cannabis Commission (ATCC) Annual Reports for fiscal years 2021 through 2025.

Data visualizations for this study were constructed using Tableau Public and are available interactively at → View Tableau Dashboard: The visualization dashboard includes five analytical charts: cigarette revenue trends, OTP growth patterns, tax rate differentials, year-over-year changes, and monthly revenue heatmaps. Static versions of key figures are reproduced in this paper.

Analytical Approach

The analysis proceeds through descriptive and comparative methods. Revenue trends for each state are examined separately, then placed in direct comparison. Key policy events are overlaid on revenue trajectories to assess their fiscal effects. Virginia's sustained revenue decline in the absence of policy changes provides a baseline estimate of secular consumption-driven revenue decline, against which Maryland's policy-driven trajectory can be assessed.

FINDINGS

Virginia: Sustained Revenue Decline Under a Stable Policy Regime

Virginia's tobacco tax revenue trajectory over the study period tells a clear and consistent story: sustained, accelerating decline in cigarette tax revenue against a backdrop of stable policy. As shown in Figure 1 and Table 1, Virginia's cigarette tax collections peaked at $234.2 million in FY2021 and have fallen every subsequent year, reaching $155.8 million in FY2025 — a cumulative decline of $78.4 million, or 33.5 percent, over four years.

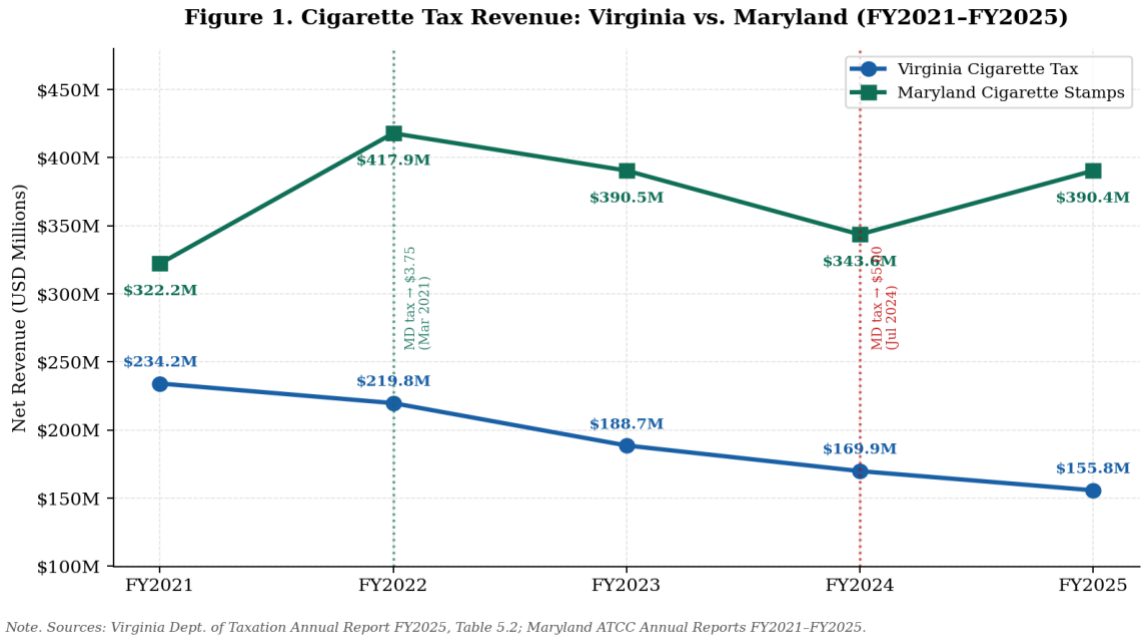

Figure 1. *Cigarette Tax Revenue: Virginia vs. Maryland (FY2021–FY2025). Source: Virginia Department of Taxation Annual Report FY2025, Table 5.2; Maryland ATCC Annual Reports FY2021–FY2025.*

Table 1

Virginia Tobacco Tax Revenue Collections, FY2021–FY2025

Fiscal Year	Cigarette Tax ($M)	OTP Tax ($M)	Total ($M)	YoY Δ Cig
FY2021	$234.2	$52.4	**$286.6**	—
FY2022	$219.8	$58.8	**$278.6**	-6.1%
FY2023	$188.7	$57.4	**$246.1**	**-14.2%**
FY2024	$169.9	$55.1	**$225.0**	-10.0%
FY2025	$155.8	$55.0	**$210.8**	-8.3%

Note. Source: Virginia Department of Taxation Annual Report FY2025, Table 5.2.

OTP revenue in Virginia remained comparatively stable, rising from $52.4 million in FY2021 to $55.0 million in FY2025 — a net increase of 4.9 percent. This divergence between cigarette and OTP trends provides strong evidence of product substitution as shown in Figure 2.

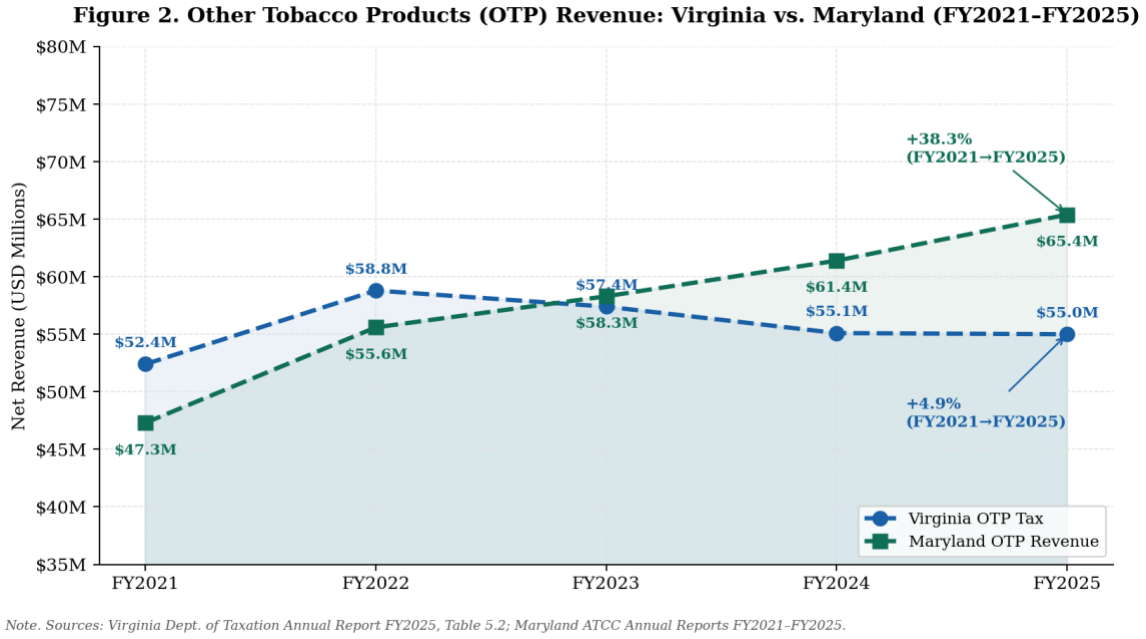

Figure 2. *Other Tobacco Products (OTP) Revenue: Virginia vs. Maryland (FY2021–FY2025). Maryland OTP grew 38.3% while Virginia grew 4.9%, confirming product substitution trend in both states.*

Maryland: Tax-Driven Revenue Volatility

Maryland's tobacco tax revenue trajectory is substantially more volatile than Virginia's, reflecting the multiple major policy interventions over the study period. As shown in Table 2, cigarette stamp revenues rose sharply in FY2022 following the March 2021 tax increase, declined through FY2023 and FY2024, and partially recovered in FY2025.

Table 2

Maryland Tobacco Tax Revenue Collections, FY2021–FY2025

Fiscal Year	Cig. Stamps ($M)	OTP Net ($M)	Total ($M)	YoY Δ Cig
FY2021	$322.2	$47.3	**$369.5**	—
FY2022	$417.9	$55.6	**$473.5**	**+29.7%**
FY2023	$390.5	$58.3	**$448.8**	-6.6%
FY2024	$343.6	$61.4	**$405.0**	**-12.0%**
FY2025	$390.4	$65.4	**$455.8**	+14.0%

Note. Source: Maryland ATCC Annual Reports FY2021–FY2025.

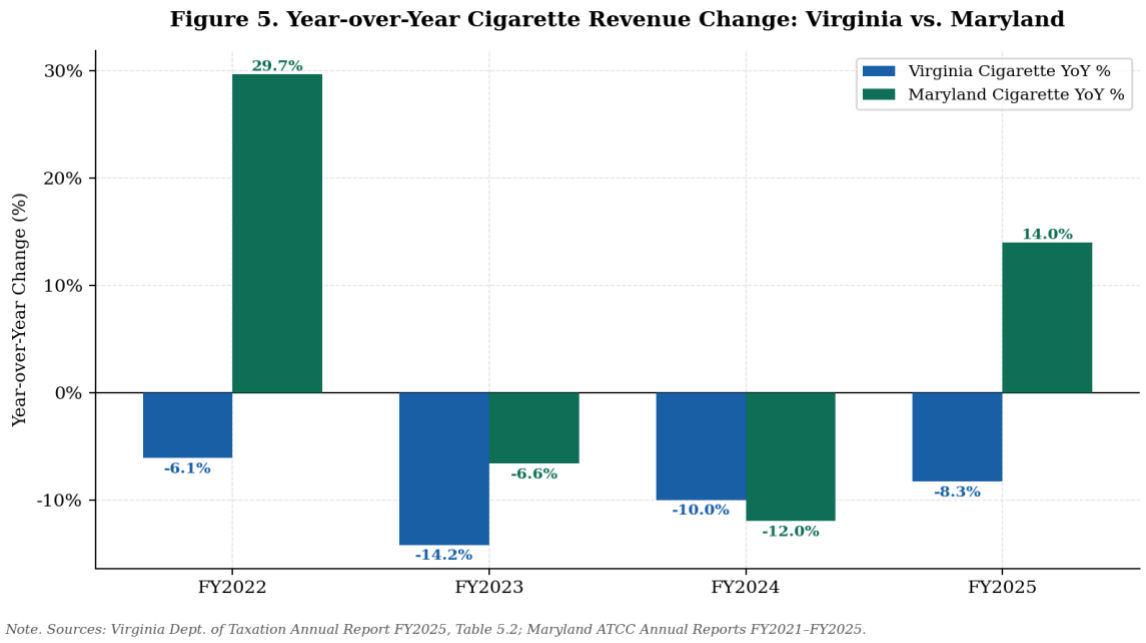

Figure 3. *Year-over-Year Cigarette Revenue Change: Virginia vs. Maryland (FY2022–FY2025). Maryland's +29.7% surge in FY2022 reflects the mechanical effect of the $3.75/pack rate increase. Virginia declined every year.*

The Virginia-Maryland Tax Differential and Cross-Border Effects

By FY2024 and FY2025, the tax differential had grown to $4.40 per pack — meaning that a Maryland consumer could save $44.00 per carton by purchasing in Virginia rather than Maryland. Figure 4 illustrates this growing divergence.

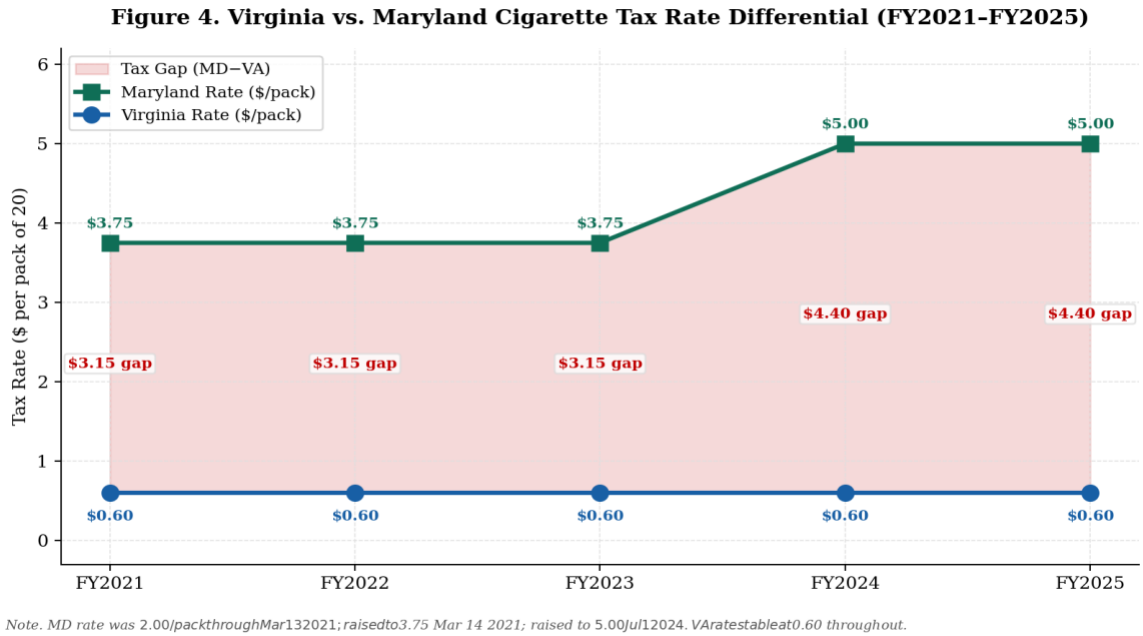

Figure 4. *Cigarette Tax Rate Differential: Virginia vs. Maryland (FY2021–FY2025). Virginia maintained $0.60/pack throughout. Maryland rose from $3.75 to $5.00 in July 2024, creating an $4.40/pack arbitrage gap.*

Table 3

Virginia-Maryland Cigarette Tax Rate Differential, FY2021–FY2025

Fiscal Year	VA Rate ($/pack)	MD Rate ($/pack)	Differential	MD/VA Ratio
FY2021	$0.60	$3.75	**$3.15**	6.25x
FY2022	$0.60	$3.75	**$3.15**	6.25x
FY2023	$0.60	$3.75	**$3.15**	6.25x
FY2024	$0.60	$5.00	**$4.40**	**8.33x**
FY2025	$0.60	$5.00	**$4.40**	**8.33x**

Note. MD rate: $2.00 through Mar 13, 2021; $3.75 Mar 14, 2021 – Jun 30, 2024; $5.00 effective Jul 1, 2024.

Maryland Monthly Revenue Patterns

Monthly Maryland cigarette stamp data reveals important within-year patterns that annual figures obscure. Figure 5 presents a heatmap of monthly revenue collections across the five-year study period.

Maryland Monthly Cigarette Revenue Heatmap (FY2021–FY2025)

Month	Fiscal Year FY2021	FY2022	FY2023	FY2024	FY2025
July	$1.2M	$30.5M	$35.3M	$36.2M	$39.9M
August	$25.3M	$37.9M	$35.4M	$27.8M	$33.2M
September	$25.1M	$36.3M	$33.0M	$33.0M	$30.2M
October	$31.4M	$42.3M	$37.8M	$32.7M	$37.6M
November	$20.0M	$37.8M	$34.6M	$27.9M	$30.8M
December	$30.5M	$32.4M	$27.9M	$27.8M	$27.0M
January	$23.2M	$34.3M	$30.6M	$23.2M	$28.0M
February	$19.8M	$29.8M	$28.0M	$29.7M	$34.5M
March	$24.8M	$34.2M	$29.2M	$22.2M	$24.6M
April	$25.8M	$35.4M	$29.5M	$28.7M	$31.0M
May	$41.4M	$31.2M	$28.0M	$33.2M	$34.1M
June	$53.8M	$35.9M	$41.1M	$21.3M	$39.7M

Figure 5. *Maryland Monthly Cigarette Stamp Revenue Heatmap (FY2021–FY2025). July 2021 anomaly ($1.2M) reflects floor tax transition. October 2022 was the highest-revenue October in the dataset, immediately preceding the DC flavor ban enforcement period.*

DISCUSSION

The Revenue Sustainability of Tax Escalation Strategies

The Maryland experience over the study period raises important questions about the long-term fiscal sustainability of tobacco tax escalation as a revenue strategy. While the March 2021 tax increase produced an immediate and substantial revenue surge of 29.7 percent, the revenue gains proved temporary. Within two fiscal years, cigarette stamp revenue had declined significantly, and by FY2024 had fallen to its lowest point in the study period despite a higher tax rate.

This pattern is consistent with the established literature on tobacco tax elasticity: an initial revenue gain from the rate increase is gradually eroded by volume reductions as consumers respond to higher prices by cutting consumption, switching products, or purchasing in lower-tax jurisdictions. The magnitude of the erosion in Maryland's case suggests that the cross-border

purchasing effect — driven by the large Virginia-Maryland differential — is a significant contributor, over and above the secular consumption decline that Virginia's trajectory suggests is occurring regardless of tax policy.

Product Substitution as a National Policy Challenge

Perhaps the most significant finding of this study, from a national policy perspective, is the consistent growth of OTP revenue across both states regardless of their different cigarette tax regimes. Maryland's OTP revenue grew 38.3 percent over the study period while Virginia's grew 4.9 percent. The parallel growth in both states — regardless of dramatically different cigarette tax regimes — reflects a fundamental structural shift in the tobacco and nicotine product market. Consumers are not quitting; they are substituting toward products that face different tax treatment.

Implications for Small Tobacco Retail Businesses

The sustained multi-year revenue decline — 33.5 percent in Virginia, with significant volatility in Maryland — translates into reduced sales volume, thinner margins, and in some cases business closure for retailers who depend on tobacco sales. Maryland's regulatory environment compounds the fiscal pressure through floor taxes, licensing requirements, and ongoing flavor ban proposals. Virginia's stable policy environment provides greater predictability even as the underlying market shrinks.

CONCLUSION

This study has examined five years of tobacco tax revenue data from Virginia and Maryland to document the fiscal and economic consequences of divergent state tobacco tax policies. Virginia's experience demonstrates that cigarette tax revenues decline at 8 to 14 percent annually even without policy changes — establishing an important baseline for evaluating the incremental effects of Maryland's tax escalation strategy.

Maryland's experience demonstrates that tax escalation produces short-term revenue gains subject to significant erosion through behavioral responses including cross-border purchasing and product substitution. The consistent OTP revenue growth in both states confirms that product substitution — not quitting — is the dominant consumer response. The $4.40 per pack tax gap persisting between the two states represents a structural cross-border arbitrage that has nationally relevant implications wherever adjacent states maintain large tobacco tax differentials.

An interactive data visualization dashboard presenting all findings in this paper is publicly available at → View Tableau Dashboard: Future research should extend this analysis to additional states, incorporate consumer survey data on purchasing location decisions, and examine employment and business survival outcomes in the tobacco retail sector.

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